Towards Fluid–Structure Interaction Modelling of Vocal–Ventricular Fold Coupling in ArtiSynth Using the Lattice Boltzmann Method

Junheng Chan
Division of Linguistics and Multilingual Studies
Nanyang Technological University, Singapore

Scan for supplementary materials
bit.ly/fsi-vvfc

Motivation

Vocal–ventricular fold coupling (VVFC) refers to mechanical coupling between the vocal and ventricular folds in the larynx. Contact between the vocal and ventricular folds may change the effective mechanical loading and vibratory behaviour. Moisik and Esling's (2014) model provides hypotheses concerning frequency, damping, irregularity and the mucosal wave.

Towards an integrated VVFC model

The long-term goal is to combine:

- **3D fold mechanics**: a continuum finite-element model (FEM)
- **Both vocal and ventricular folds**: allowing their mechanical interaction
- **Spatially resolved 3D airflow**: using the lattice Boltzmann method (LBM)
- **Two-way Fluid–Structure Interaction (FSI)**: airflow drives fold movement, fold movement changes airflow

Building on existing work

- **FEM**: Yong (2020)
- **LBM Prototype**: Kummerländer (2018)
- **ArtiSynth**: Lloyd et al. (2012)

Integration Method

Model: Yong (2020) 3D vocal–ventricular FEM in ArtiSynth

Airflow grid: three coronal D2Q9 BGK slices, each comprising 314 × 402 cells at 0.05 mm spacing

Initial glottal gap: 0.98 mm, represented by approximately 20 cells

Added pressure transfer: normalise LBM pressure patterns to a chosen 100 Pa loading scale and distribute forces to FEM nodes using surface-triangle areas and normals

Added geometry feedback: recompute airflow on the recorded deformed geometry and update the surface loads

Current scope: one offline geometry-feedback update, not yet continuous moving-boundary FSI

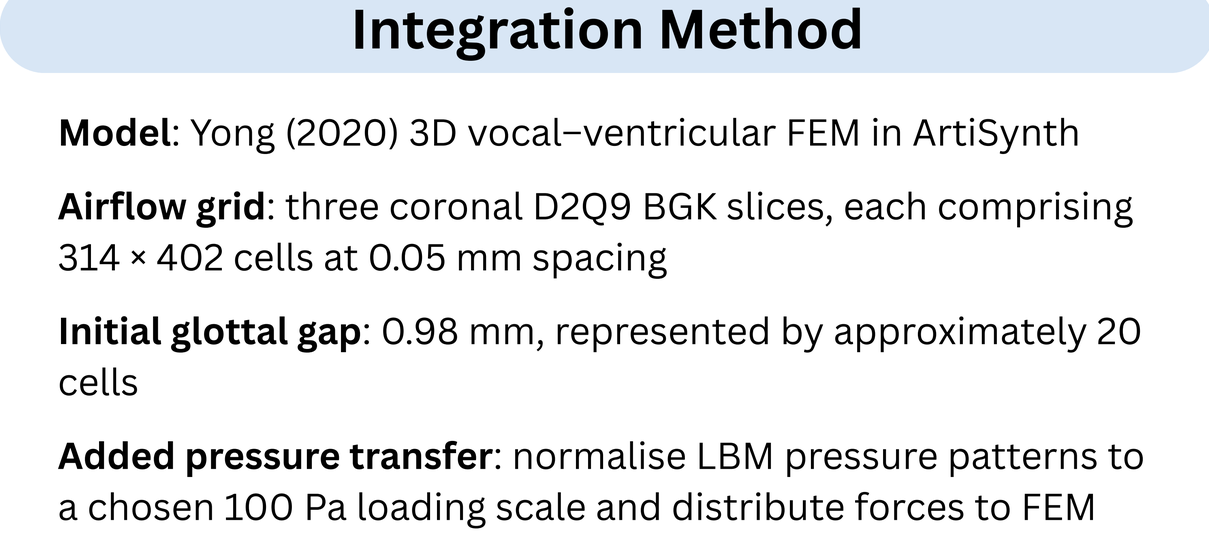


Figure 1. Mid-coronal pressure and airflow in the initial geometry. Pressure was normalised to 100 Pa for loading; speed is shown in lattice units.

Preliminary Results

Pressure loading produced FEM movement: peak nodal displacement reached approximately 0.084 mm in the vocal folds & 0.010 mm in the ventricular folds (Figure 2).

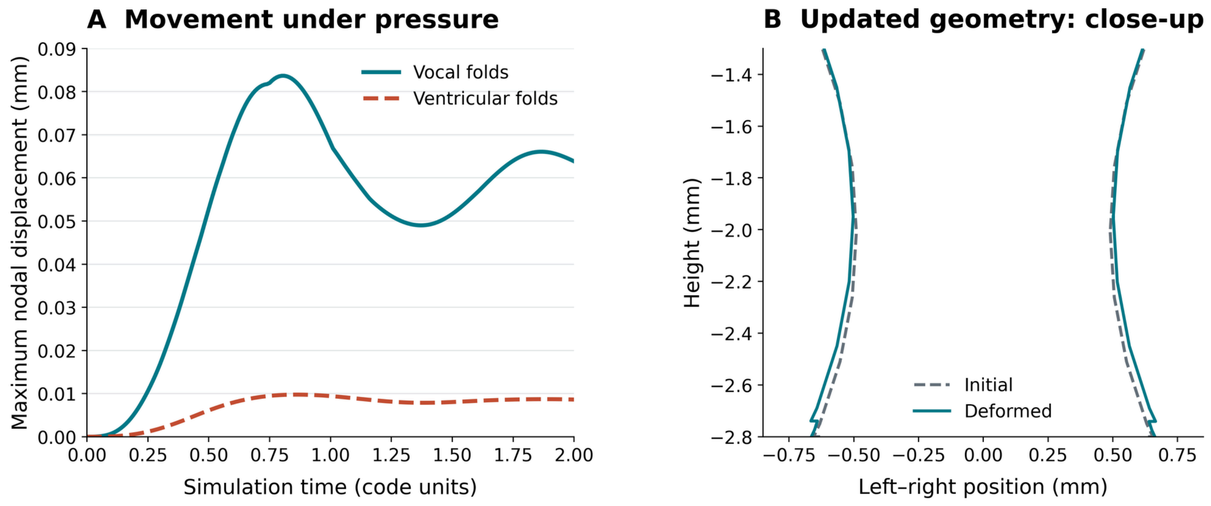


Figure 2. FEM movement under pressure loading with gravity off, and a close-up of the geometry used to recompute airflow. Displacement is not magnified.

Deformation changed the airflow: peak jet speed increased by 1.65–2.96% across the three slices at matching step counts.

Updated pressure-load resultants: increased by approximately 2.3–2.4% on the vocal folds and decreased by 4.9–5.0% on the ventricular folds (Table 1).

Table 1. Magnitudes of requested pressure-load resultants before and after the geometry update. Changes reflect both the updated pressure pattern and the deformed surface geometry. Fixed-node loads are included.

Fold	Initial load (mN)	Updated load (mN)	Change
Vocal left	11.3381	11.6063	2.37%
Vocal right	11.3391	11.6048	2.34%
Ventricular left	0.3695	0.3509	−5.02%
Ventricular right	0.3693	0.3511	−4.92%

Towards two-way FSI

Contribution: a staged proof of concept towards an FSI method for investigating VVFC. Exploratory two-way coupling was also tested (see QR supplement).

Current limitations: Developing airflow, a chosen loading scale, and near-contact resolution and stability limits in the coupled runs.

Next steps: Stabilise and validate moving-boundary coupling, calibrate physical parameters and extend airflow to 3D

Long-term goal: Self-sustained vocal–ventricular oscillation

References
Kummerländer, A. (2018, December 22). Fun with compute shaders and fluid dynamics. https://blog.kummerlaender.eu/article/fun_with_compute_shaders_and_fluid_dynamics
Lloyd, J. E., Stavness, I., & Fels, S. (2012). ArtiSynth: A fast interactive biomechanical modeling toolkit combining multibody and finite element simulation. In Soft Tissue Biomechanical Modeling for Computer Assisted Surgery (pp. 355–394). Springer.
Moisik, S. R., & Esling, J. H. (2014). Modeling the Biomechanical Influence of Epilaryngeal Stricture on the Vocal Folds: A Low-Dimensional Model of Vocal–Ventricular Fold Coupling. Journal of Speech, Language, and Hearing Research, 57(2), S687–S704. https://doi.org/10.1044/2014_JSLHR-S-12-0279
Yong, W. W. (2020). Studying vocal-ventricular fold coupling using finite-element modeling [B.A. thesis, Nanyang Technological University]. https://hdl.handle.net/10356/138299

Acknowledgements. We thank Scott R. Moisik and Wang Li-Lian for their supervision and project guidance, Scott R. Moisik for the earlier Java LBM implementation, and Wilson Yong for developing the original vocal–ventricular fold FEM.